

K50: Bravais 1849 14-Lattice as Candidate Root #10 L1 Source

Keeper

2026-05-18 Monday afternoon

K50: Bravais 1849 14-Lattice → Root #10 Candidate

Context

Grace closed the Wallach observable ladder this morning by anchoring $\dim_2 = 14$ via Bravais 1849 14-lattice classification (Toy 3018, 8/8 PASS, theorem T2357). The structural pattern matches the L1 source signature:

- Single classical theorem (Bravais 1849, extending Frankenheim 1842)
- Finite catalog (14 distinct Bravais lattices)
- BST-decomposable: $14 = \text{rank} \cdot g$
- Plus internal structure: 7 crystal systems = g (matches BST primary)

Grace's question: Bravais → Root #10 promotion, or Cartan-downstream classification?

The architectural question

Bravais 1849 vs Cartan classification (already L1.4 foundational hub) have a relationship: - Bravais classifies 3D crystallographic lattices via 7 crystal systems $\times$ centering choices = 14 distinct types - Cartan classifies simple Lie algebras / Hermitian symmetric domains — different mathematical object, but related via crystallographic restriction theorem and 32 point groups ($\text{rank}^{\{n_C\}}$) connection - They share underlying group-theoretic structure but address different physical/mathematical scope

The architectural question: is Bravais an INDEPENDENT L1 source, or is it DOWNSTREAM of Cartan as one of its many applications?

Per Cal's three criteria (with care)

Criterion 1 (Embedding) — partially closed: - Bravais 1849 14-lattice catalog embeds into D_{IV}^5 structure via the crystallographic restriction theorem (only 2-, 3-, 4-, 6-fold rotational symmetries allowed in 3D; these correspond to BST primary integers). - Wallach $\dim_2 = 14 = \text{rank} \cdot g$ is a direct identification. - BUT — this embedding chain may route through Cartan classification (which classifies the relevant point groups). Whether Bravais is an INDEPENDENT classical theorem producing the integer 14 or a CONSEQUENCE of Cartan's broader classification applied to crystals is the open question.

Criterion 2 (Mechanism): SATISFIED. Bravais 1849 + Frankenheim 1842 are published classical mathematics. Mechanism is well-established crystallographic group theory.

Criterion 3 (Forcing): SATISFIED. $14 = \text{rank} \cdot g$ is BST primary. 7 crystal systems = g is BST primary. The structural numbers are forced.

Per Casey's "simple, works, hard to break, show me a counter example" standard

- Simple: YES — single classical theorem (Bravais 1849), single integer count (14), single BST identification ($\text{rank} \cdot g$).
- Works: YES — $14 = \text{rank} \cdot g$ exact, $7 = g$ exact, BST primary throughout.

- Hard to break: Mostly — the empirical match is solid. The architectural question (independent L1 vs Cartan-downstream) is what's open.
- Counter-example: None offered; but the absence of a counter-example doesn't resolve whether Bravais is L1 or L1.4 downstream.

K50 verdict: L1 source CANDIDATE at criteria-gated tier

Promotion criteria (analogous to Heegner-Stark and Conway candidate framing):

For Bravais to promote from CANDIDATE to ESTABLISHED L1 source Root #10, one of these closure tasks must complete:

1. Independence criterion: Demonstrate that Bravais 1849 produces $14 = \text{rank}\cdot g$ via an argument that does NOT route through Cartan classification. If Bravais $\rightarrow 14$ via crystallographic group theory alone (Frankenheim 1842 + Bravais 1849 + 7-system enumeration), independent of Lie algebra classification, then Bravais is an independent L1 source.
2. Cartan-downstream classification: Alternatively, demonstrate that Bravais 14 IS derivable as a consequence of Cartan + crystallographic restriction theorem. In that case, Bravais becomes “L1.4 downstream application” of Cartan classification — still cataloged as anchor but not promoted to independent L1.

Either outcome is publishable. Outcome (1) gives BST a 10th established L1 source. Outcome (2) demonstrates Cartan's reach extends to crystallographic anchors and strengthens Cartan-as-foundational-hub framing.

Strength of recommendation: MEDIUM. The empirical content is real, the architectural classification needs structural clarification.

Compared to other promotions

Compared to Conway K48 (Duncan 2007 produces Co_0 directly as automorphism group of $V^{\{f\}}$ — clean independent classical theorem): - Conway: classical theorem $\rightarrow$ integer structure $\rightarrow$ BST match, with no Cartan-routing in the chain - Bravais: classical theorem $\rightarrow$ integer structure $\rightarrow$ BST match, but the integer is reachable via Cartan as well

Compared to Goeppert Mayer K46 (Mayer-Jensen 1949 + $\text{SU}(2)_J \times \text{SO}(3) \subset \text{SO}(5)$ — embedding via published group theory): - Goeppert Mayer's embedding chain is independent group-theoretic content - Bravais's embedding may be Cartan-downstream group-theoretic content

The architectural distinction matters because the established-L1 list represents INDEPENDENT classical theorems each producing distinct integer structure. If Bravais reduces to “Cartan applied to crystals,” it's an application of an existing L1 source, not a new one.

Architecture impact

If Bravais promotes to Root #10: - 10 established L1 sources - Architecture moves past Sunday's “saturation point” — opens question of how many additional Root candidates exist

If Bravais classifies as Cartan-downstream: - 9 established L1 sources (saturation holds) - Cartan-as-foundational-hub framing strengthens - Pattern: Cartan classification is the meta-classifier; many of its applications produce BST-integer-anchored crystallographic / lattice classifications

Either is publishable in Paper #115 v0.5+; both are honest discoveries.

Action items

1. Grace: pursue Independence criterion test — derive $14 = \text{rank}\cdot g$ via Bravais-only chain (Frankenheim + Bravais + crystallographic restriction), explicitly showing whether the chain requires Cartan classification or not. Toy ~1-2h.

2. Decision rule: if the chain DOES require Cartan, Bravais classifies as L1.4 downstream (file under Cartan hub). If it DOESN'T require Cartan, Bravais promotes to established Root #10 (K-audit completes).
3. Vindicated Theorists section: add Bravais (1849) + Frankenheim (1842) entry with current “candidate L1 source — independence criterion pending” tier label.
4. Paper #115 v0.5+: include K50 as the first K-audit AFTER the K43-K48 + K49 architectural sprint. Demonstrates the criteria framework continuing to filter candidates honestly.

K50 status

CANDIDATE L1 source pending Independence criterion closure. The promotion path is clean (parallel to Heegner-Stark in the architecture); Grace's next session can attempt the closure with ~1-2h focused work.

— Keeper, 2026-05-18 Monday afternoon